



Testing VPC Connectivity

P

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```
[ec2-user@ip-10-0-0-79 ~]$ curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />

<title>NextWork - Host a Website on Amazon S3</title>
<meta content="Let's host your very own website on Amazon S3!" name="description" />
<meta
  content="NextWork - Host a Website on Amazon S3"
  property="og:title"
/>
<meta
  content="Let's host your very own website on Amazon S3!"
  property="og:description"
/>
<meta
  content="NextWork - Host a Website on Amazon S3"
  property="twitter:title"
/>
<meta
  content="Let's host your very own website on Amazon S3!"
  property="twitter:description"
/>

<meta content="/static/og-project.png" property="og:image" />
<meta content="/static/og-project.png" property="twitter:image" />
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated virtual network within the AWS cloud that you completely control. Think of it as your own traditional data center hosted on AWS. It gives you full control over your network environment, including assigning IP address ranges, creating Subnets, and configuring Route Tables and Gateways. It is incredibly useful because it allows you to launch your AWS resources (like EC2 instances and databases) into a secure, isolated environment while managing inbound and outbound traffic on a granular level using layered security measures like Security Groups and Network ACLs.

How I used Amazon VPC in this project

I used Amazon VPC to create a secure, logically isolated network environment for my cloud resources. Specifically, I used it to host a Public Server and a Private Server within separate subnets. By configuring the VPC's Route Tables, Internet Gateways, Network ACLs, and Security Groups, I was able to successfully test, manage, and troubleshoot how these servers communicate with each other internally, as well as how they connect to the external public internet.

One thing I didn't expect in this project was...

One thing I didn't expect was how layered and granular the security settings needed to be just to allow a simple ping command to work. I learned that having a correctly configured Route Table isn't enough; both the Network ACLs (acting as the firewall at the subnet level) and the Security Groups (acting as the firewall at the instance level) must explicitly allow ICMP traffic for the servers to successfully communicate.



This project took me...

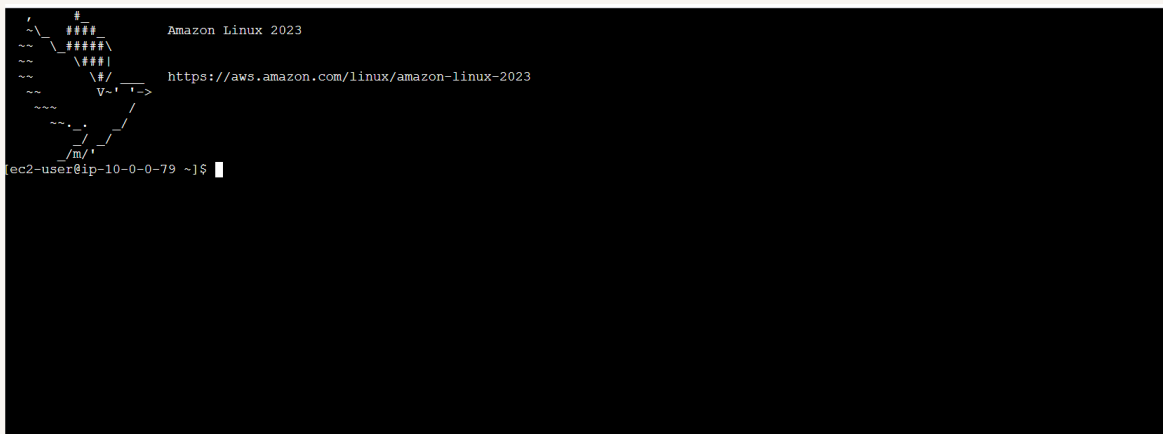
The project took me about 60 minutes to complete. This included setting up the initial VPC resources, launching the EC2 instances, and carefully troubleshooting the security rules to successfully establish connectivity.



Connecting to an EC2 Instance

Connectivity refers to how well different parts of your network communicate with each other and with external networks. It is what allows data to flow smoothly across your network, acting as the reliable backbone for any system that depends on network interactions—whether for simple web hosting or complex, large-scale operations like global streaming platforms.

In this step, we were trying to connect to the NextWork Public Server using EC2 Instance Connect via the AWS Management Console.



i-0980dd8b01976ba79 (nextwork public server)

PublicIPs: 3.121.235.14 PrivateIPs: 10.0.0.79



EC2 Instance Connect

EC2 Instance Connect is a fast and secure way to get direct SSH access to your EC2 instances using the AWS Management Console. Instead of manually generating, storing, and managing your own SSH key pairs, EC2 Instance Connect handles the complexity for you. It temporarily generates a one-time-use SSH key pair, automatically links the public key to your instance, and deletes it after a short period (typically 60 seconds) once the connection is established.

The first connection attempt failed because the NextWork Public Server's Security Group was only configured to allow inbound HTTP traffic. We were trying to connect to the server using SSH (via EC2 Instance Connect), which operates on a completely different port and traffic type. Because there was no rule allowing inbound SSH traffic, the connection was blocked by the security group.

I fixed the connection error by updating the Inbound rules of the NextWork Public Security Group. I added a new rule setting the Type to SSH (which uses Port 22) and the Source type to Anywhere-IPv4 (0.0.0.0/0). This ensured that EC2 Instance Connect could successfully initiate the SSH connection, regardless of the specific IP address it was using behind the scenes.

⊗ **Failed to connect to your instance**
Error establishing SSH connection to your instance. Try again later.



Connectivity Between Servers

Ping is a common network diagnostic tool used to check whether your computer or server can communicate with another device on a network. It works by sending small data packets (ICMP messages) to a target IP address and waiting for a reply to see how long the round trip takes. In this project, I used it to test the internal connectivity between the NextWork Public Server and the NextWork Private Server to verify that they could successfully "talk" to each other across different subnets.

I ran the command `ping [YOUR PRIVATE IPV4 DNS]` in the Public Server's terminal, replacing the bracketed text with the actual Private IPv4 address of my NextWork Private Server (for example, `ping 10.0.1.227`).

The first ping command returned only a single line and did not show any continuous replies. This meant that the Public Server successfully sent the initial ping message out, but it was not receiving any responses back from the Private Server. This indicated a connection issue—specifically, that the Private Server's network settings were blocking the incoming ICMP (ping) traffic by default.

[illegible]



Troubleshooting Connectivity

I resolved this error by updating the network security layers to allow ICMP traffic to pass through. First, I edited the NextWork Private Subnet's Network ACL by adding an Inbound and an Outbound rule (Rule number 100) to allow "All ICMP - IPv4" traffic from the public subnet's IP range. Next, because security groups act as a final firewall at the instance level, I edited the NextWork Private Security Group's Inbound rules to allow "All ICMP - IPv4" traffic coming specifically from the NextWork Public Security Group.

```
[ec2-user@ip-10-0-0-79 ~]$ ping 10.0.0.79
PING 10.0.0.79 (10.0.0.79) 56(84) bytes of data.
64 bytes from 10.0.0.79: icmp_seq=1 ttl=127 time=0.024 ms
64 bytes from 10.0.0.79: icmp_seq=2 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=3 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=4 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=5 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=6 ttl=127 time=0.029 ms
64 bytes from 10.0.0.79: icmp_seq=7 ttl=127 time=0.029 ms
64 bytes from 10.0.0.79: icmp_seq=8 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=9 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=10 ttl=127 time=0.029 ms
64 bytes from 10.0.0.79: icmp_seq=11 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=12 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=13 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=14 ttl=127 time=0.031 ms
64 bytes from 10.0.0.79: icmp_seq=15 ttl=127 time=0.033 ms
64 bytes from 10.0.0.79: icmp_seq=16 ttl=127 time=0.034 ms
64 bytes from 10.0.0.79: icmp_seq=17 ttl=127 time=0.032 ms
64 bytes from 10.0.0.79: icmp_seq=18 ttl=127 time=0.034 ms
64 bytes from 10.0.0.79: icmp_seq=19 ttl=127 time=0.032 ms
64 bytes from 10.0.0.79: icmp_seq=20 ttl=127 time=0.030 ms
```




Connectivity to the Internet

curl is a command-line tool used to test network connectivity and transfer data to or from a server. It allows you to send HTTP requests directly from your terminal to web servers and retrieve their responses, such as downloading raw HTML code, fetching data, or even uploading files.

I used curl to test the outbound internet connectivity of my VPC. By using this command on my Public Server, I was able to verify that my Public Route Table and Internet Gateway were correctly set up, allowing resources inside my Public Subnet to successfully reach and communicate with the external public internet.

Ping vs Curl

While both are tools used to test network connections, they have different primary functions. ping is used simply to check if one computer can reach another by sending a tiny message (an ICMP packet) and measuring how long it takes to get a reply. curl, on the other hand, goes a step further; it actually interacts with the destination server to transfer data, allowing you to fetch web pages, download files, or upload data using protocols like HTTP.



Connectivity to the Internet

I successfully ran a few commands. I started with `curl example.com`, which returned a basic HTML response. Ultimately, I ran `curl [https://learn.nextwork.org/projects/aws-host-a-website-on-s3]` (`https://learn.nextwork.org/projects/aws-host-a-website-on-s3`), which successfully fetched the complete and detailed HTML content of the NextWork web app directly into my terminal.

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